

640×640×3

Conv 0

320×320×16

PConv 1

160×160×32

C3k2 2

160×160×64

Conv 3

80×80×64

C3k2 4

80×80×128

Conv 5

40×40×128

C3k2 6

40×40×128

Conv 7

20×20×256

C3k2 8

20×20×256

SPPF 9

20×20×256

C2PSA 10

20×20×256

(a) PConv

X

Pad

(3,0,1,0) → 1×3 / W_h

(0,3,0,1) → 1×3 / W_h

(0,1,3,0) → 3×1 / W_v

(1,0,0,3) → 3×1 / W_v

C

2×2 Conv

s=1, C=32

Y

s=2, C=8; shared directional kernels

(b) P_2 fusion

B_2 (2)

T_3 (16) → Up ×2 → C → C3k2 → N_2: P_2 → Detect

N_2: P_2 → 3×3, s=2 → C → C3k2 → N_3: P_3

160×160×32

C3k2 19

P2

Detect

stride=4

160×160×128

Concat 18

160×160×64

Upsample 17

80×80×64

C3k2 16

80×80×256

Concat 15

80×80×128

Upsample 14

40×40×128

C3k2 13

40×40×384

Concat 12

40×40×256

Upsample 11

Conv 20

80×80×32

Concat 21

80×80×96

C3k2 22

P3

Detect

stride=8

80×80×64

Conv 23

40×40×64

Concat 24

40×40×192

C3k2 25

P4

Detect

stride=16

40×40×128

Conv 26

20×20×128

Concat 27

20×20×384

C3k2 28

P5

Detect

stride=32

20×20×256